

Effects of hillslope position on soil water flow paths in the forested hilly andesitic critical zone of Eastern China

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ABSTRACT

Soil water flow paths play a crucial role in governing the dynamics of precipitation storage within a watershed. However, the heterogeneity of critical zones complicates the flow paths of soil water in forested hilly areas. To clarify the complexity, this study introduces double-ring dye tracer experiments within a prototypical natural forested hilly catchment. Advanced image processing techniques were equipped with the dye experiment to visualize water flows intuitively at the upslope and downslope. Long-series consecutive monitoring data of soil water content (SWC) at different depths of the sites were simultaneously collected from 34 frequency domain reflectometry sensors. Based on principal component analysis (PCA) and pathway analysis, results showed significantly greater number and branching of preferential flows in the upslope than that in the downslope, which was a consequence of soil thickness and root distribution. Dye tracer experiments demonstrated that the dye within the upslope infiltrated the deeper soil layers, which indicated moisture in the upslope facilitated groundwater recharge. Inversely, the dye within the downslope infiltrated the shallow soil layers, which suggested moisture in the downslope supplemented the shallow soil. The predominant flow type in the upslope was macropore flow with low interaction, whereas the downslope featured macropore flow with high interaction. The PCA results identified that soil properties, precipitation characteristics, root systems, faunal activities, and geographic locations were the main factors affecting preferential water flow paths in hillslopes. Soil layers characterized by numerous flow paths and a uniform distribution of flow path widths exhibited high SWC. Moreover, high SWC was observed soil layers either above or below the lateral flow zones. Soil water flow path and infiltration variability significantly influenced the spatial distribution of water and thereby shaped the pathways and processes involved in runoff generation in the critical zone. These findings enhanced our understanding of hydrological processes in the hilly andesitic critical and provided new insights into the mechanisms governing soil water flow and transport dynamics.

1. Introduction

In recent years, climate change has resulted in a more erratic spatial and temporal distribution of precipitation, which increased the

variability of hydrological processes, for example, runoff generation, groundwater recharge, and soil water dynamics (Seneviratne et al., 2010; Snyder et al., 2019; Wu et al., 2020; Yuan et al., 2023). Soil water flow path controls the dynamics of precipitation storage within

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watersheds, which highly influences the water, energy, and biogeochemical cycles (Seneviratne et al., 2010; McColl et al., 2017).

Soil water flow paths constitute a fundamental component of the water movement within the critical zone (Lin, 2010). The modes of soil water flow paths can be classified into soil matrix and preferential flow in terms of flow type, and vertical and lateral flow in terms of flow direction. The role of interaction between the macropore domain and the soil matrix domain is critically important for soil water storage (Weiler and Naef, 2003; Weiler and Flühler, 2004). Macropores can transport large volumes of water, which can later infiltrate the surrounding soil matrix through capillary suction. Preferential flow in macropores may bypass most of the soil's porous matrix (Beven and Germann, 1982), which limits the soil ability to store, filter, and buffer water (Clothier et al., 2008; Beven and Germann, 2013). Preferential flow has been recognized as a potentially important infiltration mechanism in soils, which may increase the leaching potential of soils and thereby accelerate the transport of nutrients, pesticides, and pathogens (Khne et al., 2009; Jarvis, 2020). The existence of preferential flow can greatly affect the flow production patterns and processes of surface runoff and subsurface runoff (Guo and Lin, 2018). However, for structured soils with preferential flow in macropores, the effect of a soil water saturation deficit on infiltrability and surface runoff generation can be limited by the reduced interaction between macropores and the soil matrix (Alaoui et al., 2011). Furthermore, the presence of preferential flow paths (PFP) in heterogeneous aquifers may dramatically alter aquifer behavior, which underscores the importance of accounting for PFP in solute transport simulations and environmental risk assessments (Nan et al., 2019). However, the heterogeneity of critical zones, including variations in soil physicochemical properties and plant root development, introduces complexity into PFP studies (Edwards et al., 1992; Oshun et al., 2016; Chen et al., 2018). Thus, for forest hillslopes, the quantification of the effects of macropore–matrix interactions on surface runoff and soil water storage remains unclear and poses a challenging task for studies.

In the critical zone of hilly areas, PFP facilitates the transport of water, solutes, energy, and gases to greater soil depths, which thereby reduces soil erosion (Allaire et al., 2009). PFP also can induce significant alterations in the physical, chemical, and biological characteristics of the soil (Bogner et al., 2012). Well-connected PFP can enhance soil infiltration capacity and contribute to soil structure development. These pathways play a crucial role in shaping the hydrological response to rainfall within the critical zone and subsequently influence the dynamic processes of groundwater and surface water (Jarvis, 2020). However, PFP may also pose risks, such as subsurface drainage during intense rainfall, which leads to geological hazards and groundwater pollution. During intense rainfall events, these pathways may serve as key channels for subsurface drainage, which could elevate water pressure due to rapid recharge. In some cases, this can result in subsurface water flow along slopes, disrupt the existing mechanical equilibrium, and increase the incidence of geological hazards such as landslides and mudflows (Tao et al., 2020; Wang et al., 2023). PFP offers an efficient conduit for the transport of pollutants and accelerates the contamination of groundwater.

Hillslope water flow transport is important to conduct in-depth research on flow production and transport processes in changing environments (Coles et al., 2017; Erler et al., 2019; Wine and Davison, 2019). Strengthening the understanding of soil water movement under the nonuniformity of unsaturated soil is necessary (Burt and McDonnell, 2015; McGuire and McDonnell, 2010). Investigating soil water flow modes can enhance the precision of forecasting soil moisture levels and flow production patterns (Wang et al., 2015; Sprenger et al., 2019; Ma et al., 2020). The dye tracer method is the most employed technique due to its ease of use, direct observability, and cost effectiveness (Flury and Flühler, 1994). Stump and Hendry (2012) studied the vertical and lateral movement of the shallow soil water down to a depth of 6 m and revealed the important impact of heterogeneous structures. Timbe et al. (2014) found that different land use and soil type significantly

influenced the generation and transport of the water flow. Presently, PFP studies using the dye tracer method mainly focus on the influence of various factors, for example, different vegetation types (Zhang and Yuan, 2019; Guan et al., 2023), stump heights (Liu et al., 2023), hydraulic properties (Kotlar et al., 2020), and irrigation conditions (Hopkins et al., 2016; Niels et al., 2019). The spatial variation in water infiltration capacity was attributed to the differences in soil structure among slope positions (Zhu et al., 2023). Differences in the hydraulic processes of each hillslope position contributed to the redistribution, transportation, and storage of surface and belowground water, which resulted in differing availabilities of soil water resources and utilization by plants (Chen et al., 2021). Soil moisture dynamics at different hillslope positions after rainfall varied widely due to the controls of soil properties, topography, and nonequilibrium flow (Lu et al., 2024). The hillslope is an essential natural spatial gradient that influences hydrological processes by affecting water transport, water distribution, and surface flow. However, clarity regarding the differences in preferential flow on various hillslopes within the hilly forested critical zone is currently lacking, and the mechanisms by which PFP influences soil water storage remain unclear (Liu et al., 2020; Yan et al., 2023).

Although preferential flow has been widely recognized, its variation across different hillslope positions and the underlying mechanisms linking soil structure to water transport remain insufficiently understood. This study addresses these gaps by comparing macropore flow characteristics between upslope and downslope soils through dye tracer experiments and by proposing a conceptual framework linking hillslope position, soil heterogeneity, and macropore–matrix interaction. The internal observations of the soil water flow remain challenging, and the understanding of soil water dynamics is limited due to the lack of detailed observations. A forested hilly catchment in subtropical eastern China was selected as the experimental site of forested critical zone in the lower Yangtze River basin (Zhang et al., 2020). The main purpose of this study is to gain insights into the morphological characteristics of PFP under different slopes through the visualization of dye movement within the soil. The main objectives are to 1) reveal the features and relationship of soil lateral transport path and soil water storage at different hillslope locations and 2) propose conceptual framework of soil water transport paths regarding hillslope location–soil heterogeneity–preferential flow to elucidate the influence of different soil structures and hillslope locations on the water flow. For this purpose, the double-ring dye tracer experiments equipped with advanced image processing techniques were conducted at different hillslopes. Some statistical methods such as principal component analysis (PCA) and path analysis were utilized to explore systematically the effects of hillslope positions on soil water paths, the morphological characteristics and variability attributes of PFP, and further reveal the underlying relationships between PFP and soil water storage. This work is crucial for advancing our understanding of hydrological processes within the andesitic critical zone and for providing theoretical foundation and scientific data for model simulations in the hilly forested watershed.

2. Materials and methods

2.1. Description of study area

The study was conducted in the 0.79-hectare Nandadish Experimental Catchment, which represents a typical natural forested hilly critical zone located at 32.28° N, 118.2° E. Nandadish is situated within the Chuzhou Hydrological Experimental Station, located in the Huashan Basin in eastern China (Fig. 1). Nandadish exhibits a low-hill landform with surface slopes ranging from 6.7° to 17.1° in various directions and an altitude difference of 12.9 m (Gu et al., 2018). The study site was a typical deciduous broadleaved forest stand, a tree density of 1128 trees ha⁻¹, and a basal area of 17.7 m² ha⁻¹ (Zhang et al., 2020). This landform features a brief flat summit, straight and elongated backslope, and a short flat footslope. The climate of the study site is the East Asian

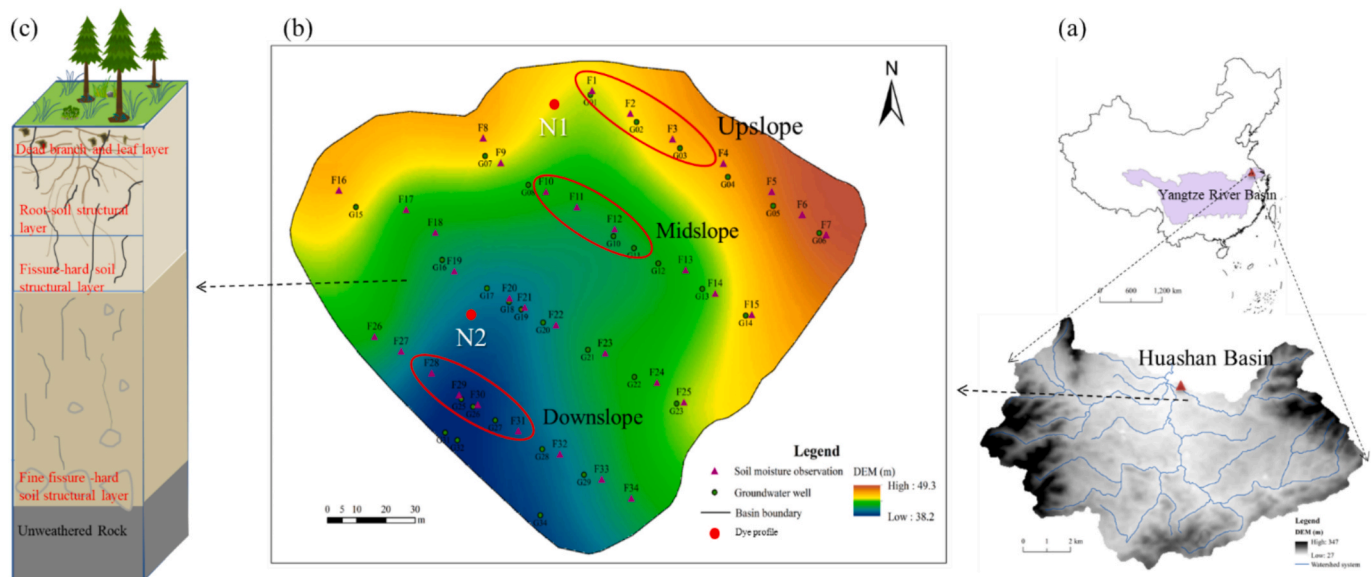


Fig. 1. Study area overview (a) and sampling site layout (b), together with soil profile distribution (c).

monsoon climate, which is traditionally categorized into a warm and wet summer monsoon with intense rainfall and a cold, dry winter monsoon characterized by light precipitation. The average annual air temperature and water surface evaporation recorded from 1951 to 2016 are 14.9 °C and 891 mm, respectively (Zhang et al., 2019). The annual precipitation in the region averages around 1007.7 mm. The soil layer thickness within Nandadish ranges from 0.5 m to 7 m (Zhang et al., 2020). Based on the soil profiles excavated, the soil profile can be broadly divided into three layers in the study area: a litter layer consisting of dead branches and leaves (0–10 cm), a root-affected soil structural layer (approximately 10–60 cm), and a compact subsoil layer with fine fissures below 60 cm (Fig. 1). Notably, the upper slope positions exhibit thicker soil layers, whereas the lower slope positions feature thinner soil layers (Gu et al., 2018). The catchment's bedrock consists of Mesozoic andesitic consolidated rock, characterized by limited fissures (Gu et al., 2018).

2.2. Experimental design

2.2.1. Soil water content and soil physicochemical properties monitoring

Thirty-four SWC observation points were evenly distributed across the catchment to examine SWC variations (Fig. 1; Liu et al., 2022). At the top, middle, and bottom of the hillslope, 3–4 SWC observation points were chosen to calculate the average SWC for the three representative positions (Fig. 1). Soil profile moisture rapid measuring instruments (SM-1, ADCON, Austria) were used to monitor volumetric SWC in the catchment at 10 cm depth intervals from 0 to 1 m with a time interval of 15 min. In an earlier work, the gravimetric soil moisture content of standard points was determined in the study area through the oven-drying method. The corresponding volumetric water contents were as calculated calibration values using bulk density values. To ensure the accuracy of the measurements further, a fitting curve ($y = 1.11x - 3.2$ ($R^2 = 0.95$)) was developed to link the measured SWC values with the calibrated values. The SWC data used in this study were calibrated using a fitting curve to maintain the accuracy of the soil moisture content measurements.

The points near the SWC observation points at the top, middle, and bottom of the hillslope were selected as representative points for determining soil physicochemical properties in the study area (Fig. 1). Soil samples were collected using a ring knife (5 cm in height and 5 cm in diameter) at depths of 0–10, 10–20, 30–40, 50–60, 60–70, and 80–90 cm. The soil particle composition, including clay (0–0.002 mm), silt

(0.002–0.02 mm), fine sand (0.02–0.2 mm), and sand (0.2–2 mm), fractions was quantified using a laser particle size analyzer (Mastersizer 2000, Malvern Instruments, Malvern, UK). Soil particle size measurements were also calibrated against instrumental standards samples. Soil bulk density and porosity were determined through the oven drying method.

Soil moisture characteristic curves were determined using the pressure plate method (DIK-3404, Japan) by applying suction pressures of 1, 3, 10, 25, 50, 101, and 1500 KPa (soil water potentials). The 1500 KPa suctions were applied for wilting point (Gu et al., 2017; Tetegan et al., 2015). The soil moisture characteristic curves were then fitted using the Van Genuchten model to obtain optimal fit parameters for each soil depth and to calculate the soil saturated moisture (Van Genuchten, 1980; Li et al., 2018). Soil organic matter was determined utilizing the potassium dichromate external heating method. Soil pH was determined using a glass electrode in suspension (water–soil ratio was 2.5:1) (Qiu et al., 2022).

2.2.2. Dye tracer experiments

Brilliant Blue (BB) FCF (C.I. 42090) dye tracer was introduced into the water to visualize the infiltration of water within the soil profile (Flury and Flühler, 1994). This tracer is well used for visualizing flow patterns in vadose zone hydrological research, but it also exhibits some deficiencies, such as problematic sorption and lower mobility than pure water in small pores (Nobles et al., 2010; Wang and Zhang, 2011). The BB FCF (C.I. 42090) dye tracer was chosen primarily because of its low toxicity and excellent visibility (Flury and Wai, 2003). A concentration of 4 g L⁻¹ ensured the dye remained visible even after dilution and adsorption (Flury and Flühler, 1995).

The dye infiltration experiment was conducted to visualize flow pathways in the upslope (N1) and downslope (N2) locations within Nandadish (Fig. 1). To ensure consistent dye penetration, a two-ring method was employed. First, a stainless-steel sample column with an inner diameter of 20 cm was inserted into the soil to a depth of 10 cm. Subsequently, another sample column with an inner diameter of 30 cm was positioned in the outer ring (Fig. 2a). Ten liters of BB solution was poured gradually and evenly over 12 h using a fixed shower unit to maintain a constant water level at approximately 2 mm/min above the soil surface, and minor fluctuations were within ± 0.5 mm. This method helped minimize the risk of artificially increasing macropore flow due to high ponding pressure. The purpose of using this amount of water was to observe the deep soil water flow path characteristics under conditions of

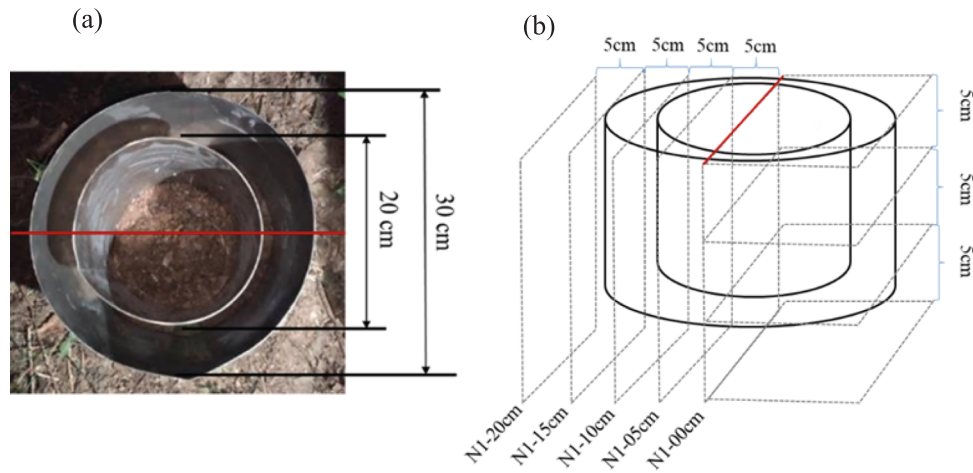


Fig. 2. Dye Tracer experiment utilizing the double ring method and staining profile diagram ((b) was a schematic representation intended to clearly show how the transverse and vertical cross sections of soil profiles are divided and not all of the excavated transects are depicted in the diagram.).

sufficient precipitation. Some pure water was added into the outer ring to facilitate the full leakage and diffusion of the dye tracer. Twenty-four hours after irrigation, five vertical profiles were excavated at distances of 0, 5, 10, 15, and 20 cm from the center of the inner ring. More than 20 transects were excavated from the ground downward at 5 cm intervals until the dye stain disappeared (Fig. 2b). The excavation depth reached soil layers with no traces of staining. In addition, a soil pit was dug in front of these profiles for photographic documentation. Photographs of the soil profiles were captured using a digital camera (Canon EOS 5D Mark III; resolution: 22 megapixels). The generated digital images had dimensions of approximately 5760×3840 pixels.

2.3. Calculations and statistical analysis

2.3.1. Image processing and optical calibration

After capturing the images, geometric and optical calibration was conducted using OpenCV and NumPy modules in Python to generate binary images. The areas in the images that were dyed were subsequently transformed to black using tone value adjustments, whereas the remainder stayed white. Dye coverage (DC), with a vertical resolution of 0.5 mm, was computed for all vertical dye patterns. The paths through which the staining reagent flowed (black) were assigned a value of 1, whereas the areas not traversed (white) were designated as 0. The image information matrix was generated using MATLAB programming.

The following common preferential flow indicators can be derived through the analysis of the stained images:

- (1) DC ratio (%) is the proportion of the total stained area to the entire area of the soil profile (Flury and Flühler, 1994). DC is calculated using the following formula:

$$DC = \left(\frac{D}{D + ND} \right) \times 100\% \quad (1)$$

where D (cm^2) represents the stained area, and ND (cm^2) represents the unstained area.

- (2) Depth of soil matrix flow (UniFr) is the depth at which uniform infiltration occurs, typically when DC exceeds 80 %, as defined by Mei et al. (2018).
- (3) Preferential flow ratio (PF-fr) is the percentage of the total stained area, as expressed by the following formula:

$$PF - fr = 100 \bullet \left(1 - \frac{Unifr \bullet W}{TSA} \right) \quad (2)$$

where PF-fr is the preferential flow ratio (%), W is the width of the stained soil area (20 cm in this study), and TSA represents the total stained area.

- (4) Length index (LI) is the summation of the absolute differences between the depth-stained area ratios of consecutive soil layers within a stained profile, as expressed by the following formula:

$$LI = \sum_{i=1}^n |DC_{i+1} - DC_i| \quad (3)$$

- (5) Staining paths are characterized as individual independent or consecutive multiple stained pixels within each row of a stained image. Staining path number (SPN) and the width of staining paths serve as indicators of the connectivity and branching of preferential flow.
- (6) Stained path width (SPW, mm) represents the width of individual staining paths within a specific soil layer depth, as calculated using the following formula:

$$SPW = \frac{SN}{R} \quad (4)$$

where SN indicates the number of pixels occupied by independent staining paths, and R is the image resolution ($0.5 \text{ pixel} \bullet \text{mm}^{-1}$ in this study).

Following the framework proposed by Weiler and Flühler (2004), five different flow path types can be categorized based on the SPW

Table 1

Classification of preferential flow types based on stained path width (SPW) categories

Flow type	SPW 1 < 20 mm	SPW 2 > 200 mm
Macropore flow with low interaction with surrounding matrix	>50 %	<20 %
Macropore flow with mixed interaction (high and low) 20–50	20–50 %	<20 %
Macropore flow with high interaction	<30 %	<20 %
Heterogeneous matrix flow and fingering	<20 %	30–60 %
Homogeneous matrix flow	<20 %	>60 %

Note: The proportion was calculated by dividing SPW by DC. adapted from Weiler and Flühler (2004).

(Table 1).

2.3.2. Statistical analysis and plotting

Data preprocessing was completed with Microsoft Excel 2016, whereas statistical analyses were performed using SPSS. For PCA, a stepwise multiple linear regression model and pathway analysis, this study utilized SPSS Version 25.0 (IBM, USA). Path analysis facilitated the quantitative separation of direct and indirect influence on soil water transport caused by the selected factors. Graphs and plots were generated using R software, Origin 2020, and PowerPoint. Statistical significance was assessed at the 0.05 significance level.

3. Results

3.1. Soil physical and chemical characteristics

Physical and chemical properties of the soil at various depths were determined at three different locations of the experimental catchment. The soil structure exhibited significant spatial heterogeneity (Table 2). The soil was low in organic matter except for the topsoil, and the overall content of clay grains was high in the study area (Table 2). Therefore, the soil structure primarily consisted of blocky, prismatic units that featured well-developed horizontal and vertical fissures. Higher organic matter content in the upslope was mainly distributed in the 0–20 cm soil layer, whereas for the midslope and downslope, 0–10 cm soil layer showed the highest organic matter content (Table 2). The soil in the study area exhibited high clay content, with a decreasing trend from shallow to deep soil layers. The average clay particle content among different locations of the slope showed the following hierarchy: midslope > upslope > downslope (Table 2). The maximum difference in mean clay content between the midslope and downslope was only 3 %, which indicated practically the same content. Soil pH slightly varied across locations and depths, ranging from 6.04 to 6.89 (Table 2). The highest porosity was consistently observed in the 0–10 cm soil depth, whereas the lowest porosity was detected in the 50–60 cm soil depth in the upslope and downslope, and in the 30–40 cm soil depth in the midslope. The porosity in the 0–10 cm soil layer was 6 % higher for the upslope compared with the midslope and downslope, which was indeed a significant difference. The pattern of soil bulk density in the study area was inverse to the soil porosity pattern. In the study area, soil of the upslope exhibited the wilting point ranging from 0.04 % to 0.36 % and saturated moisture conductivities (SMC) ranging from 0.49 % to 0.70 %. Soil of the midslope displayed the wilting point ranging from 0.01 % to 0.04 % and SMC ranging from 0.39 % to 0.70 %. Soil of the downslope had the wilting point varying from 0.01 % to 0.23 % and SMC ranging from 0.42 % to 0.70 %.

3.2. Infiltration path analysis through dye experiment

3.2.1. Developmental characteristics of water transport pathways in soils

The dye area of the soil layer slightly increased in the upslope from 5 cm to 10 cm depth (dye area: 85.79–94.97 cm²), which suggested that lateral flow was not well developed in that depth (Fig. 3a). The dye area of the soil layers at 10–20 and 50–60 cm depths showed an increasing trend (Fig. 3a), which indicated the development of lateral flow in these depths. The dye area of the 30 cm layer in the upslope reached the maximum peak (611.78 cm²), which indicated that the lateral flow was well developed near the 30 cm soil depth in the upslope. The dye area of the soil layer at other soil depths (30–50 cm, below 60 cm) in the upslope decreased with the increasing depths, which suggested that vertical flow dominated the soil layer at these depths (Fig. 3a). The dye area at a depth of 5–10 cm in the downslope increased from 53.35 cm² to 236.02 cm² (Fig. 3a), which indicated that the lateral flow was well developed at that soil depth. The dye area of the downslope slightly increased in from 10 cm to 20 cm depth, and the dye area slightly decreased from 20 cm to 40 cm depth, which indicated that vertical flow

Table 2
Physical and chemical properties of the soil at different depths and locations in the NDD.

Plot	Depth (cm)	Particle size distribution			Sand (0.02–0.2 mm) (%)	Sand (0.2–2 mm) (%)	Organic (g/kg)	Organic carbon (g/kg)	pH	Porosity (%)	Bulk density (g/cm ³)	Moisture content	
		Clay (0–0.002 mm) (%)	Silt (0.002–0.02 mm) (%)	Silt (0.002–0.02 mm) (%)								Wilting point (cm ³ /cm ³)	Saturated moisture content (cm ³ /cm ³)
N1 Upslope	0 ~ 10	36.10	43.27		20.17	0.47	24.55	14.24	6.98	46.81	1.41	36	70
	10 ~ 20	30.62	36.66		32.38	0.34	13.93	8.08	6.28	44.66	1.47	10	55
	30 ~ 40	26.61	32.62		38.70	2.07	8.21	4.76	6.43	44.46	1.65	4	65
	50 ~ 60	27.70	33.98		36.37	1.95	7.91	4.59	6.35	37.51	1.66	6	60
	60 ~ 70	27.74	34.30		36.57	1.39	7.81	4.53	6.38	40.69	1.57	7	50
	80 ~ 90	25.02	32.54		40.61	1.84	4.10	2.38	6.42	37.34	1.66	6	49
N2 (Midslope)	0 ~ 10	32.53	37.83		27.40	2.24	19.81	11.49	6.04	40.83	1.57	1	55
	10 ~ 20	30.54	35.04		33.89	0.53	7.10	4.12	6.04	40.77	1.57	3	39
	30 ~ 40	30.44	34.81		33.30	1.45	4.00	2.32	6.15	35.25	1.72	1	44
	50 ~ 60	27.76	31.80		39.60	0.85	3.83	2.22	6.31	36.88	1.67	4	51
	60 ~ 70	28.63	29.35		38.94	3.07	3.33	1.93	6.33	43.97	1.48	4	51
	80 ~ 90	28.96	29.25		34.02	7.77	2.53	1.47	6.22	39.60	1.60	3	45
N3 (Downslope)	0 ~ 10	35.72	35.66		25.41	3.21	19.72	11.44	6.28	39.17	1.61	23	7
	10 ~ 20	36.72	36.31		24.38	2.59	9.43	5.47	6.45	41.12	1.56	2	45
	30 ~ 40	25.86	34.23		39.10	0.82	5.15	2.99	6.55	38.45	1.63	3	48
	50 ~ 60	22.44	29.17		45.96	2.43	3.97	2.30	6.43	36.52	1.68	1	52
	60 ~ 70	20.71	28.25		44.51	6.53	3.76	2.18	6.46	37.76	1.65	5	42
	80 ~ 90	19.37	29.84		48.30	2.49	3.72	2.16	6.57	40.53	1.58	3	40

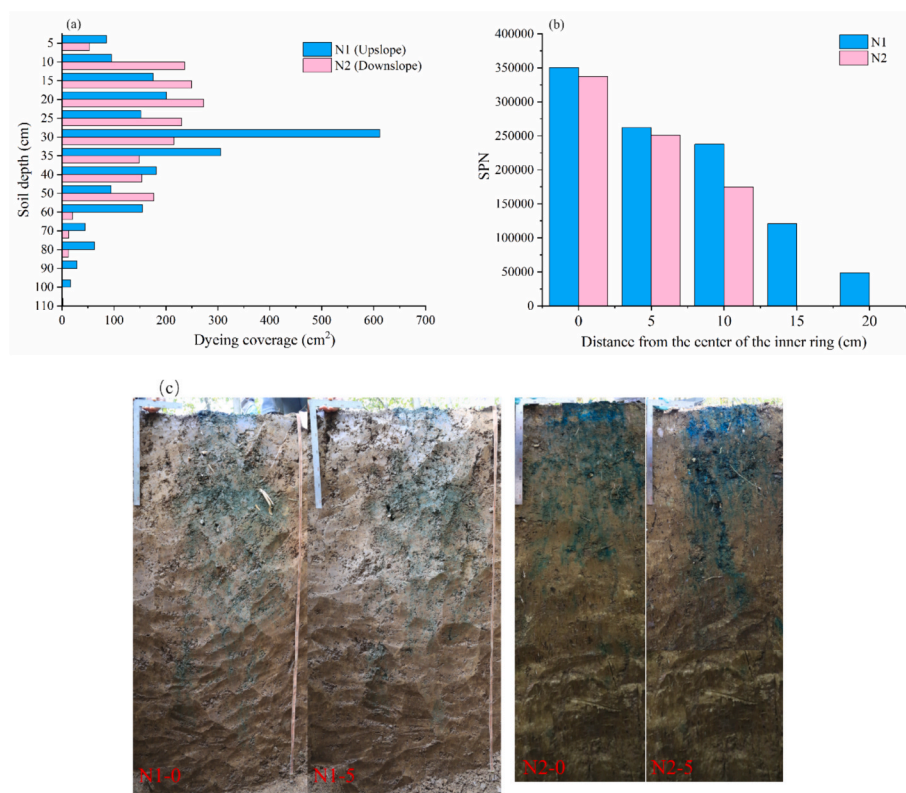


Fig. 3. (a) Cross-sectional dye area at different soil depths at the upslope and the downslope; (b) The stained path numbers (SPN) of vertical profiles with different distance from the center of the inner ring at the upslope and the downslope; (c) the vertical cross-section of the distribution of the infiltrated water (N1-0 represent the central vertical profile, and N1-5 represent the profile with 5-cm distance from the center).

was dominant in the soil layer from 10 cm to 40 cm depth, with some development of lateral flow. The dye area of the soil layers from 40 cm to 50 cm depths increased from 153.66 cm² to 176.72 cm² in the

downslope (Fig. 3a), which indicated the development of lateral flow in that depth. After lateral flow occurred at 50 cm depth, the N2 dye profile infiltrated only to 80 cm depth (Fig. 3a).

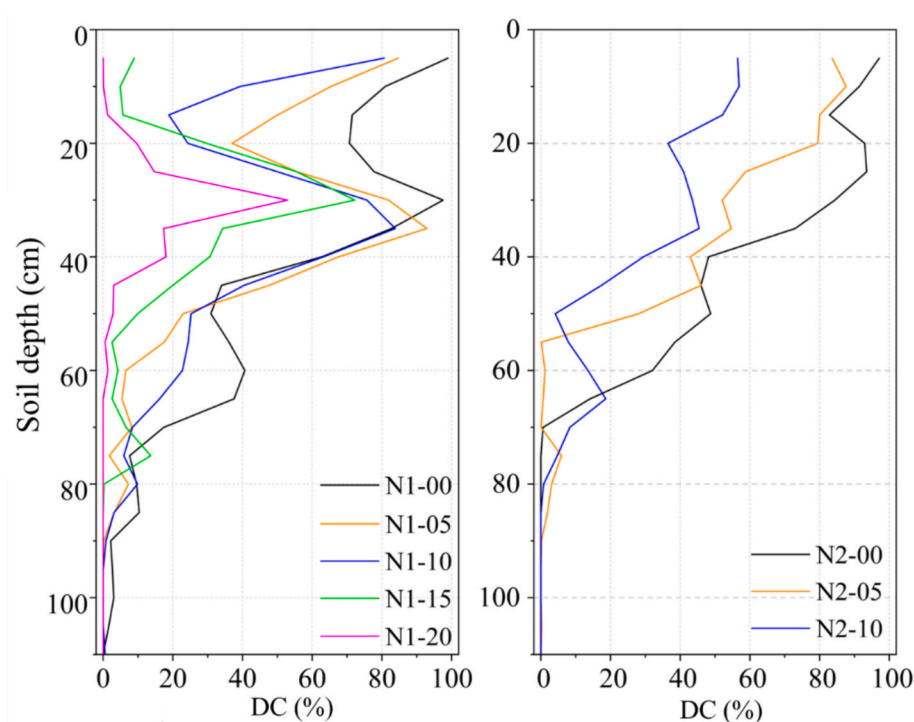


Fig. 4. Soil dye coverage for different profiles at different slope (N1 refers to the upslope position and N2 refers to the downslope position; the numbers represent the distance from the central profile).

The SPNs in the upslope were mainly concentrated within 0–15 cm from the center of the inner ring (95.22 %), and the SPNs in the downslope were mainly concentrated within 0–10 cm from the center of the inner ring (100 %), as shown in Fig. 3b, which indicated that the preferential flow of the soil layer in the upslope was larger branched than that in the downslope (Fig. 3c). The SPNs in the upslope were 1.34 times higher than those in the downslope (Fig. 3b), which showed that the number of preferential flows in the upslope was significantly larger than that in the downslope. The SPNs in the upslope were slightly larger than the SPNs in the downslope within 0–5 cm from the center of the inner ring, whereas the SPNs in the upslope were much larger than the SPNs in the downslope within 10–20 cm from the center of the inner ring, which suggested that the water flow paths in the soil in the downslope were weakly diffused, and vice versa in the upslope.

3.2.2. Infiltration path analysis

The DC showed an overall decreasing trend with the increasing depths in all upslope dye profiles, except for the 20–35 cm soil depths. DC decreased abruptly at 0–20 cm and below 35 cm depth within all the upslope dye profiles, and increased at depths of 20–35 cm. The DC of all upslope dye profiles fluctuated with the increasing soil depths (Fig. 4), which indicated the heterogeneity of soils in the upslope. For all the downslope dye profiles, DC showed an overall decreasing trend with the increasing depths. In addition, DC decreased with the increasing distance from the center dye profile in the soil layer above 50 cm ($N2-10 < N2-5 < N2-0$), and the soil layer below 50 cm was not characterized by this pattern (Fig. 4).

The maximum depths of infiltration for the five dye vertical cross sections of soil profiles ranged from 77 cm to 114 cm in the upslope, the deepest infiltration occurred at N1-10, and the shallowest depth of infiltration was at N1-15 (Table 3). The maximum depth of infiltration for the dye profiles ranged from 66 cm to 102 cm in the downslope, and the deepest infiltration profile occurred at N2-5. The maximum depth (114 cm) of dye infiltration in the upslope was deeper than that (102 cm) in the downslope, which indicated that the preferential flow could carry water deeper into the soil of the upslope than into soil of the downslope. The total SPNs in the upslope was much larger than those in the downslope (Table 3), which indicated more transport paths were in the upslope and the transport was more extensive in the upslope than in the downslope. The LIs of the vertical cross sections of soil profiles ranged from 327 % to 452.7 %, which illustrated the high variability of the local preferential flow. The wide range of depth of soil matrix flow and small PF-fr in the downslope, and the narrow range of soil matrix flow and large PF-fr in the upslope, indicated that the preferential flow in the upslope was highly developed, and the soil matrix flow in the downslope

was highly developed. In summary, the number and branching of preferential flows in the upslope were significantly greater than those in the downslope. Dye within the upslope infiltrated deeper soil layers and contributed to groundwater recharge.

3.2.3. Distribution characteristics of the SPNs and SPWs

N1-0, N1-05, N2-0, and N2-05 were selected as representative profiles for the analysis of SPN and SPW in the study. Within N1-0 above 50 cm depth, the 10–20 cm soil layer exhibited the highest SPN, followed by the 30–40 cm soil layer. The SPWs of the primary water transport path showed a gradual variation, ranging from 50–100 mm, 20–50 mm, to 10–20 mm, within the 0–50 cm depth of N1-0 (Fig. 5). Below 50 cm in N1-0, SPN decreased with the increasing soil depth, while the SPW gradually expanded. The distribution characteristics of the SPNs in N1-05 was consistent with that of N1-0. Above 50 cm in N1-05, the distribution of various SPWs was even, and it was dominated by SPWs of 100–200 mm below 50 cm. The N2 profiles exhibited fewer SPNs compared with the N1 profiles. Above 50 cm in N2-0, SPWs of 20–50 and 50–100 mm served as the main channels for water transport (Fig. 5). Below 50 cm in N2-0, wide paths dominated the SPWs. The SPN and SPW patterns in N2-05 aligned with those of N2-0, and the distinction was a deeper dye path in N2-05.

Flow type classification, according to Weiler and Flühler (2004), indicated the presence of homogeneous matrix flow down to a minimum depth of 5 cm across all hillslopes (Fig. 6). In N1-0, the predominant flow type was low-interaction macrospore flow (42.28 %), characterized by fine, independent paths, mainly found at depths of 40–65 and 10–25 cm in the soil profile. This flow type was followed by macrospore flow with high interaction (33.64 % of the profile), primarily concentrated at depths of 0–10, 25–35, and below 60 cm (Fig. 6). Other dye vertical cross-sections of soil profiles of the upslope were predominantly characterized by macrospore flow with high interaction and macrospore flow with low interaction. In the downslope profiles, macrospore flow with high interaction served as the primary flow type, followed by macrospore flow with mixed interaction. The central vertical profile in the downslope exhibited greater consistency with the other profiles in terms of flow path type and depth distribution. By contrast, the upslope showed a higher degree of variability in flow path types. In the upslope, the predominant flow type was macrospore flow with low interaction, whereas the downslope featured macrospore flow with high interaction.

3.3. Impact of soil physicochemical characteristics on water transport

In this investigation, DC, soil physicochemical properties, various SPWs, and soil depth were selected as key indicators. PCA was applied to extract four primary factors denoted as F1, F2, F3, and F4 (Table 4). F1 explained 52.64 % of the variance, which was mainly associated with positive scores for clay, silt, organic matter, and field water holding capacity, while F1 featured negative scores for sand and bulk density. F1 primarily reflected the soil physicochemical properties of the study area. F2 was positively correlated with DC, porosity, and SMC. F2 was indicative of the dye tracer amount (representing the amount of rainfall), which was also influenced by soil porosity and SMC. The combined variance explained by F1 and F2 reached 82.69 %. F3 displayed positive scores for SPW within the ranges of <5, 5–10, 10–20, and 20–50 mm. Conversely, it exhibited a negative score for staining path widths of 100–200 mm. F3 primarily reflected the count of small-pore SPW, likely influenced by the fine pores. F4 was positively correlated with SPW within the range of 50–100 mm, which indicated that F4 represented the count of larger-pore SPWs, primarily influenced by broader elements such as insects and root formations of pores. Overall, the primary factors influencing the primary water flow pathways in the study area included soil physicochemical properties, fundamental rainfall characteristics, the influence of fine pores, and the count of larger-pore SPWs. The stepwise multiple linear regression model for the DC was developed to determine the influence of soil physicochemical characteristics on the

Table 3
Preferential flow characterization parameters.

Site	Depth of infiltration (cm)	UniFr (cm)	SPN	PF-fr (%)	LI (%)
N1-00	106	0–7, 24–35	350,605	61.21	420.26
N1-05	108	0–3, 29–36	262,297	66.45	398.63
N1-10	114	0–4, 29–34	237,635	91.58	452.7
N1-15	77	/	120,598	/	327.63
N1-20	95	/	48,671	/	246.55
N2-00	66	0–29	337,311	31.22	275.71
N2-05	102	0–12, 17–19	250,597	39.34	375.64
N2-10	76	/	174,671	/	447.28

Note: N1 is an upslope position and N2 is a downslope position; the meaning of two numbers in column UniFr is two discontinuous depths.

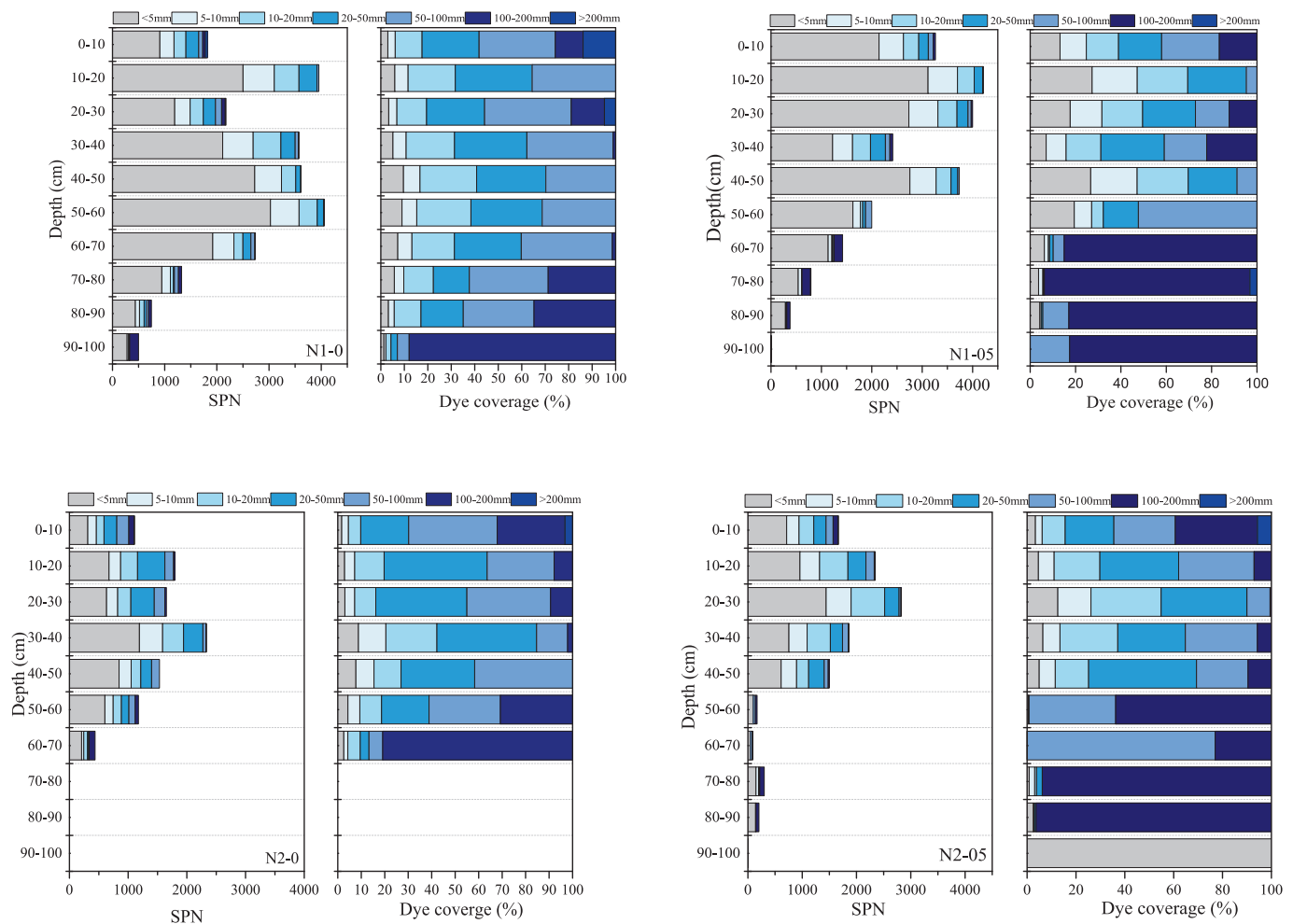


Fig. 5. Distribution of different widths of staining path numbers (SPN) and their dye coverage (DC) widths in the horizontal soil profile of a representative dye profiles (N1-0, N1-5, N2-0, and N2-5) (N1 refers to the upslope position and N2 refers to the downslope position).

DC:

$$DC = 4.0 * (Silt) - 36.5 * (Saturation\ moisture\ content) - 28.7 * (Wilting\ point) + 2.3 * (Porosity) - 1.2 * (Depth) + 230.1 * (Bulk\ density) - 466.2$$

($R^2 = 0.98$, $P = 0.000 < 0.01$)

DC was significantly affected by soil physicochemical characteristics.

3.4. Characteristics of the SWC variation

The average SWC in 2021 varied across the hillslope positions: 38.68 % $\pm$ 6.8 % in the upslope, 34.23 % $\pm$ 5.5 % in the midslope, and 39.02 % $\pm$ 6.4 % in the downslope. The highest SWC was in the downslope, whereas the lowest SWC was in the midslope (Fig. 7). The 60 cm depth showed the lowest SWC of 31.35 % and could be considered the “dry layer” in the midslope. SWC in the upslope exhibited a three-phase pattern: 0–40 cm layer gradually increased from 32.79 % to 40.89 %, 40–60 cm layer decreased from 36.92 % to 40.89 %, and 60–100 cm layer increased from 36.92 % to 43.21 %. The downslope displayed three prominent peaks in SWC at different depths: 20 cm (39.62 %), 40 cm (44.10 %), and 80 cm (45.04 %). For the three positions, SWC peaks were consistently observed at 40 cm soil depth, and troughs were evident at 60 cm depth.

The highest fluctuations of SWC were observed in the 0–10 cm soil depth during the rainy season across all the three locations (Fig. 8). The variability of SWC within the 0–50 cm depth was larger than that within the 60–100 cm depth for the upslope and midslope. In the midslope, the

SWC at a depth of 60–100 cm remained stable. Conversely, in the downslope, SWC exhibited significant fluctuations from the surface down to 90 cm during the rainy season. Additionally, the variability of SWC in the rainy season was higher in the upslope and the midslope than in the dry season. The SWC pattern during the dry season was also similar to that during the rainy season in the midslope. In the downslope region, the variation of SWC from 50 cm to 90 cm depth was larger than that within the 0 to 50 cm depth during the dry season (Fig. 8).

Fig. 9 shows the SWC response to three different types of rainfall in the study area: uneven rainfall with strong instantaneous rain, continuous uniform rainfall, and heavy rainfall. During the rainfall period, the SWC was high in the 20 and 40 cm soil layers and low in the 30 cm soil layer of the upslope. The SWC in the midslope exhibited the lowest magnitude of response to precipitation, which indicated that the midslope primarily served as a channel for downward water transport with low water retention. In the deep soils (50–100 cm), the SWC in the upslope remained high even after rainfall ended, whereas the SWC in the midslope and downslope significantly decreased (Fig. 9). This outcome suggested that upslope had a higher water-holding capacity in the deep soil layers. In addition, rainfall recharged quickly to groundwater of the upslope, and the groundwater levels stayed high after the rain event compared with the downslope (Fig. 10). The water (e.g., rain) in the upslopes seeped deeper into the soil and promoted groundwater recharge.

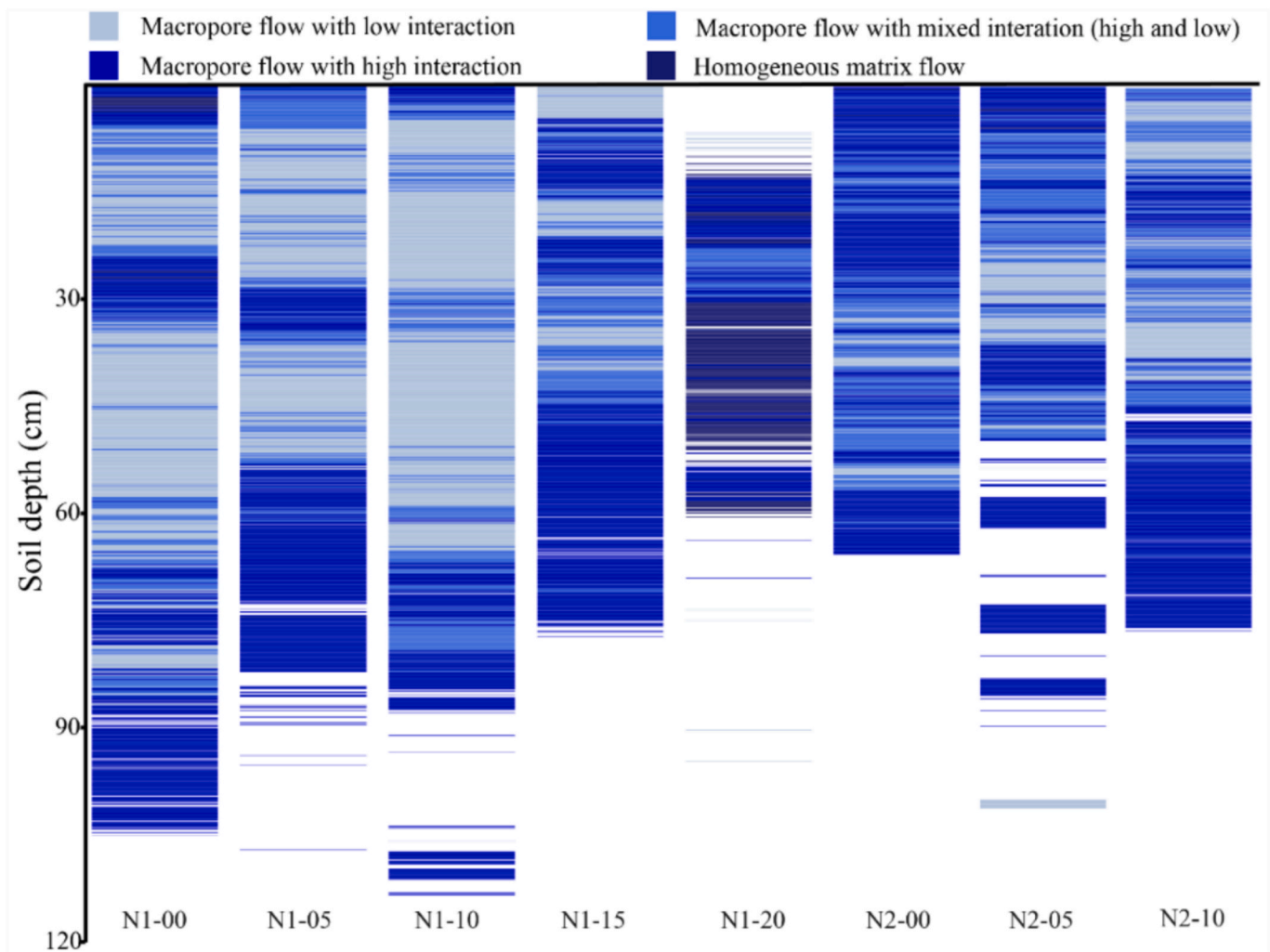


Fig. 6. Classification of flow types in soil profiles (N1-00 ~ N1-20) in the upslope and (N2-00 ~ N2-10) in the downslope using dye tracer analysis (N1 refers to the upslope position and N2 refers to the downslope position).

4. Discussion

4.1. Analysis of soil water movement pathways

The preferential flow of the soil layer in the upslope was larger branched than that in the downslope. Lateral preferential flow pathways were notably developed in the soil layer, primarily within the depth range of 10–30 cm in the upslope, particularly for the 30 cm depth (Fig. 4). This preferential flow pattern aligned with findings from previous studies by Guan et al. (2022) and Luo et al. (2019), which identified similar preferential flow phenomena in the Southwest Karst Mountains and North China Mountains. This phenomenon may be attributed to the existence of lateral pore structures formed by animal activity within the soil layer (Jačka et al., 2021). Furthermore, the presence of a well-developed vegetative root system at this layer, encompassing lateral and vertical root systems, contributed to these preferential flow patterns (Luo et al., 2019). Additionally, lateral flow within the 50–60 cm deep soil layer could be linked to animal activity within the soil. Notably, lateral flow was evident from a depth range of 5–50 cm in the downslope, with prominent development observed within the 5–10 and 40–50 cm soil depth intervals. This lateral flow behavior restricted water infiltration into deeper soil layers and limited the depth of infiltration in N2-0 to a maximum of 66 cm (Table 3). Comparing the coefficient of variation of the dye area of cross-sections in the upslope with that in the downslope revealed that the upslope

exhibited greater variability. This difference could be primarily attributed to soil heterogeneity. The thickness of the soil layer in the upslope was considerably greater and reached 7 m, whereas the thickness of the soil layer in the downslope did not exceed 1 m. The increased thickness of the soil layer provided more space for the root systems and soil fauna to move within the soil and ultimately resulted in greater spatial variability (Zhang et al., 2015). The preferential flow in the upslope soil layers was more branched than in the downslope. This finding could be attributed to differences in soil texture above and below the slope, in addition to soil thickness and root distribution. Specifically, the porosity in the 0–10 cm soil layer was over 6 % higher in the upslope than in the downslope. Moreover, the sand content in the 0–50 cm soil layer was significantly higher in the upslope than in the downslope. By contrast, soil depth greater than 50 cm in the downslope had low porosity, even though the sand content increased. The DC showed a decreasing trend with the increasing depths in the upslope dye profiles, except for 20–35 and 50–60 cm soil depths. This pattern emerged due to the presence of two lateral preferential pathways with robust water transport capabilities. By contrast, the DC in the downslope exhibited a more uniform and homogeneous pattern. This uniformity suggested that soil moisture primarily infiltrated uniformly in the form of soil matrix flow, or the morphological differentiation between soil matrix flow and preferential flow was not significantly pronounced. The flow path diffusion of soil water in the downslope was weaker than that in the upslope, and the depth of dye infiltration fluctuated at the top and bottom of the slope,

Table 4
Rotated component matrix.

Parameter	F1	F2	F3	F4
DC0	0.72	0.65	0.24	0.10
DC5	0.50	0.86	−0.01	0.08
DC10	0.37	0.91	−0.04	−0.15
DC15	−0.11	0.90	0.32	0.25
DC20	−0.32	0.91	0.22	0.11
Clay	0.99	0.07	−0.02	0.08
Silt	0.99	0.01	−0.15	0.06
Fine Sand	−0.99	−0.06	0.10	−0.02
Organic	0.98	0.18	−0.06	0.08
pH	0.77	0.20	−0.58	−0.16
Porosity	0.67	0.61	0.04	0.42
Bulk density	−0.90	0.03	−0.02	−0.44
Wilting point	0.94	0.05	−0.33	−0.03
Saturated moisture conductivity	0.65	0.66	0.11	−0.35
<5 mm	−0.10	0.09	0.95	−0.18
5 ~ 10 mm	−0.01	0.41	0.90	0.02
10 ~ 20 mm	0.01	0.69	0.70	0.02
20 ~ 50 mm	0.47	0.57	0.50	0.44
50 ~ 100 mm	0.28	0.35	−0.52	0.55
100 ~ 200 mm	0.21	−0.14	−0.91	−0.08
>200 mm	0.88	0.13	−0.43	−0.12
Soil depth	−0.80	−0.47	−0.30	−0.16
Eigenvalues	11.58	6.61	2.21	1.01
Variance Explanation Rate (%)	52.64	30.05	10.07	7.78
Cumulative variance explained (%)	52.64	82.69	92.76	97.56

Note: DC represent stained area ratio in the longitudinal profile, and numbers represent distance from internal diameter.

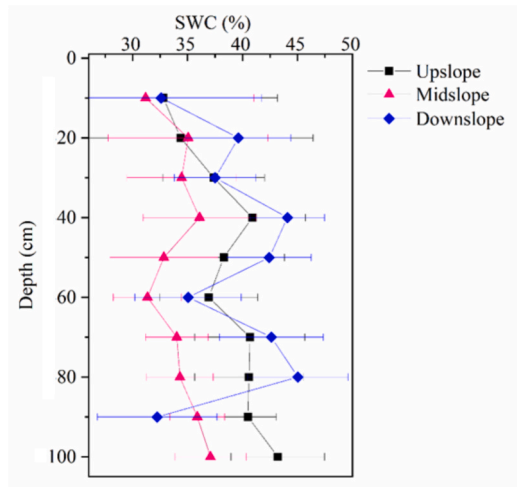


Fig. 7. Vertical distribution of mean soil water content throughout.

which was caused by the high degree of heterogeneity and the wrap-around properties brought about by the preferential flows (Wang et al., 2018; Luo et al., 2019).

Preferential flow strongly affects the water infiltration capacity and increases the complexity and messiness in the redistribution pattern of water in soils (Alaoui et al., 2011). In the study, the dyed water infiltration time significantly increased from upslope to downslope. This finding indicated that the upslope had a higher potential to infiltrate the rainfall during the rainy season, whereas the downslope had a poor drainage discharge capacity. Enhanced preferential flow and water exchange may facilitate vertical and lateral redistribution of soil moisture, and thereby contribute to diversified water sources and improved water supply across the slope (Zhu et al., 2019).

4.2. Examination of soil flow types

The primary flow type observed in the soil profiles in the upslope was

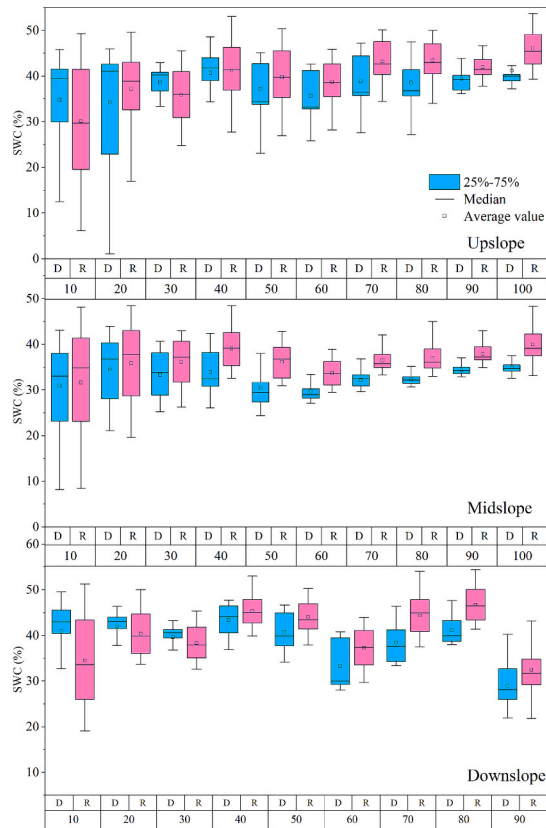


Fig. 8. Variations in soil water content (SWC) at different depths in the upslope, midslope, and downslope for the year 2021. (Note that herein D indicates the dry season, and R indicates the rainy season).

macropore flow with low interaction. This flow type exhibited fine, independent characteristics, featuring primary SPWs ranging from 0 to 5 mm, 5 mm to 10 mm, and 10 mm to 20 mm. Macropore flow with low interaction was predominantly distributed at the depths of 40–60 and 10–20 cm within the soil layers (Fig. 5). Such category of SPWs demonstrated a higher number of SPNs and a wider distribution, which facilitated extended dispersion and deeper water infiltration, and contributed to the development of fine fissures in the upslope soil layers. Conversely, macropore flow with high interaction accounted for 33.64 % in the upslope profile, which was mainly concentrated in soil layers within the 0–10, 25–35, and below 60 cm. Macropore flow with high interaction occurred in the 0–10 cm soil layer because the soil porosity was highest in this layer (Table 2), in the 25–35 cm layer because of the well-developed root system of vegetation (related to the good development of lateral flow), and in the soil layer below 60 cm because the low number of fissures became the critical channel within such depth (Alaoui et al., 2011). This type of flow was mainly associated with path widths of 20–50 and 50–100 mm. Although such flow type occupied the second-largest proportion, it was responsible for over 50 % of the water transport pathways and served as the primary conduit for water transport in the upslope of the study area. The transition from N1-05 to N1-20 experienced shifts from a prevalence of high interaction macropore flow to low interaction macropore flow and subsequently back to high interaction macropore flow. These variations were predominantly due to the significant spatial heterogeneity and wrap-around characteristics inherent to the preferential flow paths within the soil layers of the upslope.

In the downslope profiles, macropore flow with high interaction emerged as the primary mode. Staining paths with widths between 20 mm and 50 mm, and those ranging from 50 mm to 100 mm, which served as the principal channels for soil water transport. Additionally,

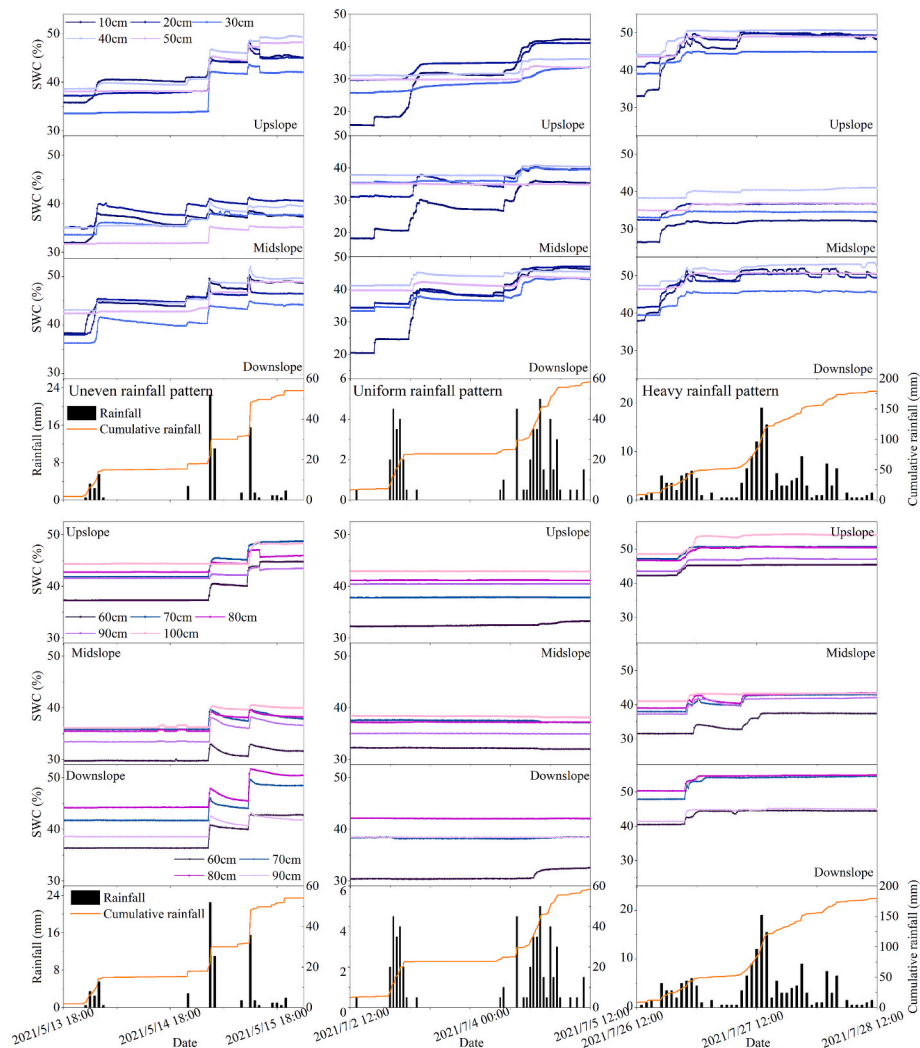


Fig. 9. Response of the SWC to rainfall events on different slopes under the three rain patterns.

macropore flow with mixed interaction accounted for 38.37 % of the soil water transport pathways, followed by macropore flow with low interaction with a minor proportion of 7.09 %. Channels with SPWs ranging from 100 mm to 200 mm played a pivotal role in soil water transport within the topsoil and deep soil layers. In general, soil water routing approaches exhibited greater width in the subslope soil layers. Notably, the soil structure may remain largely unaltered over decades without any human tillage intervention, particularly for the macropores and biopores of the soil.

The reasons for the difference between the upslope flow with low interaction and downslope macropore flow with high interaction can be explained as follows. Macropore flow with low interaction in the upslope mainly occurred in areas where lateral flow was not well developed because the direction of pore development aligned well with the direction of infiltration, which allowed rapid passage of water through macropores with good connectivity, but with limited exchange capacity with the surrounding soil matrix. By contrast, macropore flow in the downslope was irregularly distributed macropores and varying orientations. These macropores exhibited a higher capacity for exchange with the surrounding soil matrix, although this capacity diminished with depth (Fig. 3c). Macropore flow with high interaction dominated water movement upslope and downslope, and showed strong transport capacity (Beven and Germann, 2013). Unlike low-interaction flow that bypassed soil matrix, high-interaction macropores actively exchanged water with surrounding soil, and enhanced infiltration depth and lateral

spread. This exchange was driven by capillary forces and pressure differences at macropore boundaries. Vegetation promoted these pathways as dense roots, and biological activity created connected biopores, especially downslope (Jačka et al., 2021; Luo et al., 2019). During intense or uneven rainfall, macropores quickly transmitted water exceeding the soil matrix capacity. High-interaction macropores efficiently redistributed this water, reduced runoff, and improved soil moisture.

4.3. Major factors affecting soil water transport pathways

PCA highlighted the significance of soil physicochemical properties, precipitation characteristics, the influence of fine pores, and the count of larger-pore SPWs in determining soil water transport pathways in the study area. Among them, soil physical and chemical properties explained 52.64 % of the variance. According to the stepwise multiple linear regression model, the main soil physicochemical properties that played important roles were silt, porosity, wilting point, bulk density, and SMC, except soil depth. Path analysis was introduced to understand further the specific elements of soil physicochemical properties that influence soil water movement pathways. Fig. 11 shows the relationship between the DC of the central stained profile and soil physicochemical properties within the study area. The primary factors affecting the DC varied across soil layers, where soil depth had a direct path coefficient of -0.96 . Soil depth had a direct, important effect on soil moisture

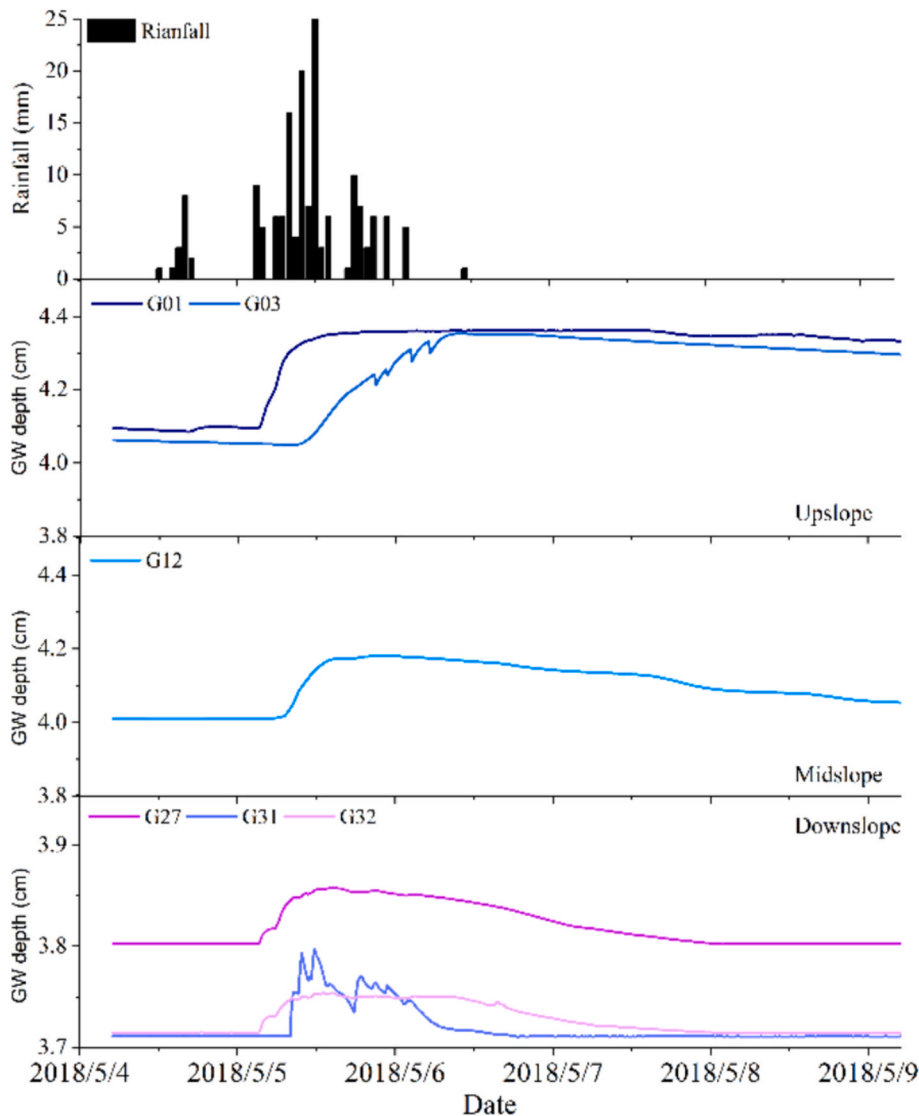


Fig. 10. Response of the groundwater (GW) to a rainfall event on different slopes.

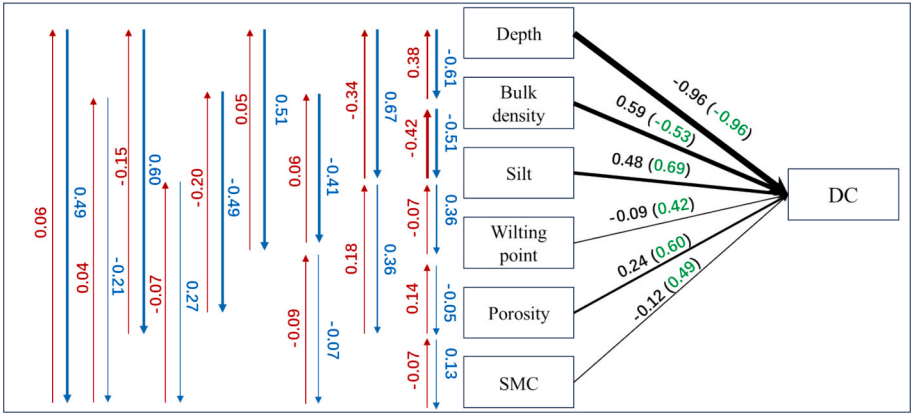


Fig. 11. Direct and indirect path coefficients of key soil properties on the DC in central stained profiles (The green number represented correlation coefficient; The black number represented direct path coefficient; The red and blue number represented indirect path coefficient.).

transport. Other soil properties, for example, organic matter, porosity, wilting point, and SMC also decreased from the topsoil layer to the deep soil layer (Table 2). Therefore, soil depth was a comprehensive index to

reflect soil properties (Table 5, Fig. 11). The DC increased with bulk density, and silt content increased. The DC decreased as porosity increased (Fig. 11) possibly because the greater the porosity (smaller

Table 5

Direct and indirect path coefficients of key soil properties on the DC in central stained profiles.

Variable	Correlation coefficient	Direct path coefficient	Indirect path coefficient					
			Depth	Bulk density	Silt	Wilting point	Porosity	SMC
Depth	−0.96	−0.96	/	0.38	−0.34	0.05	−0.15	0.06
Bulk density	−0.53	0.59	−0.61	/	−0.42	0.06	−0.20	0.04
Silt	0.69	0.48	0.67	−0.51	/	−0.07	0.18	−0.07
Wilting point	0.42	−0.09	0.51	−0.41	0.36	/	0.14	−0.09
Porosity	0.60	0.24	0.60	−0.49	0.36	−0.05	/	−0.07
SMC	0.49	−0.12	0.49	−0.21	0.27	−0.07	0.13	/

Note: SMC indicates saturated moisture content.

capacity), the higher the infiltration rate, which resulted in a smaller DC. Soil depth, silt content, porosity, bulk density, and wilting point directly influenced the soil water transport paths (see Table 5). Moreover, these factors indirectly influenced the soil water transport paths by impacting other environmental factors, as presented by the direct and indirect path coefficients (Fig. 11). Hence, not merely a single soil property, but a combination of soil properties (Alaoui et al., 2011), including soil depth, silt content, porosity, bulk density, and wilting point, influenced soil water transport pathways. Precipitation characteristics such as the amount, duration, and intensity of rainfall are other important factors influencing soil water transport pathways. The amount of rainfall directly affected the infiltration depth of soil water, and the duration and intensity of rainfall collectively affected the rate of soil water transport (Vidon and Cuadra, 2010; Zhu et al., 2014). Small-pore SPWs were likely influenced by fine pores, which resulted from soil temperature fluctuations and small root systems. Larger-pore SPWs were primarily influenced by insects and root formations of pores (Mei et al., 2018; Zhao et al., 2020). Both were preferential flow paths and directly acted on the transport of soil water. The formation of lateral flow paths in the upslope and downslope was critically related to the growth patterns of vegetation roots and the activities of soil fauna. In summary, soil depth had a direct, important effect on soil water transport and other soil properties, and a combination of soil properties, precipitation characteristics, root systems, faunal activities, and slope positions shaped the preferential water flow paths in hilly critical zones.

4.4. Relationship between SWC and soil water flow paths

4.4.1. Factors influencing SWC variations

The highest mean SWC was observed in the downslope, whereas the lowest was found in the midslope within the study area (Figs. 7 and 9). This difference may be attributed to the downslope acting as a moisture convergence zone, whereas soil water in the midslope was more susceptible to transport and less prone to storage (Figs. 9 and 10). A notable SWC peak was noted at a soil depth of 40 cm in the three representative locations (Figs. 7 and 9), which indicated that 40 cm deep soil layer likely served as a key reservoir for water storage during the infiltration within the study area. Notably, the SWC at a depth of 80 cm in the downslope exhibited the highest values in the deep soil layers, possibly influenced by groundwater dynamics.

The largest variation of SWC were observed in the uppermost 10 cm of soil, which demonstrated the topsoil's strong response to precipitation and surface soil evaporation (Liu et al., 2022). During the rainy season, the variability in SWC in the upslope and midslope was larger than that during the dry season within the study area. This phenomenon can be attributed to the subtropical monsoon climate, characterized by intense precipitation, and increased temperatures and evaporation rates during the rainy season. SWC exhibited minimal variation between soil depths of 50 and 100 cm in the midslope (Figs. 8 and 9), which further illustrated that the hilly midslope was more susceptible to transport soil water and less prone to storage water. The 0–50 cm soil layer in the upslope showed the largest variation in SWC whereas the 60–100 cm soil layer in the downslope presented the greatest variation because

numerous flow paths were observed in the 0–50 cm soil layer of the upslope, and the 60–100 cm soil layers in the downslope were highly heterogeneous.

4.4.2. Implications for soil water transport processes in hilly andesitic critical zones

The comprehensive findings of this work indicated that vertical matrix flow predominated in the 0–10 cm soil layers of the upslope, whereas lateral flow developed from 10 cm to 30 cm, particularly near the 30 cm depth where the development of lateral flow was vigorous (Fig. 12). Trees affected macropore content and connectivity directly by their specific root system (channels around roots; Jačka et al., 2021). High SWC at 40 cm depth may be attributed to the pronounced development of lateral flow in the soil layer. Given the predominant macropore flow with low interaction in this region (Fig. 12), a soil water storage layer could form at the depth of 40 cm. Moreover, the 40 cm soil layer in the upslope was characterized by multiple staining paths and a uniform distribution of various SPWs (Fig. 5), which was more conducive to soil water storage. The reduced SWC at 60 cm depth may be attributed to this soil layer's transitional nature, which shifted from low-interaction to high-interaction macropore flow (Fig. 12). The 60 cm soil layer showed well-developed lateral flow and a sudden increase in SPWs (Fig. 5), which likely facilitated soil water transport downward. For soil layers in the downslope, high SWC was noted at a depth of 20 cm likely due to the predominance of mixed macropore flow in this layer depth, characterized by a large number of SPNs and a uniform SPW distribution (Figs. 5 and 12), which enhanced the storage of soil water. The high SWC at 40 cm depth in the downslope may be attributed to its positioning above the region with developed lateral flow, and this soil layer was also characterized by macropore flow with low interaction, which facilitated soil water storage. The high SWC within 70–80 cm depth range may be influenced by groundwater (Liu et al., 2022). The SWC in the upslope remained at a high SWC (50–100 cm) after rainfall and decreased in the midslope and downslope (Fig. 9) due to the variability of soil water flow path and permeability. The upslope had a greater number of deep soil paths, most of which were low-interaction macropore flows. Overall, a soil layer characterized by numerous flow paths with a uniform distribution of flow path widths and dominance of macropore flow with low interaction exhibited higher SWC. Soil layers positioned above or below areas of developed lateral flow could also lead to higher SWC.

Generally, numerous soil water flow pathways were observed in the upslope soil layers, which exhibited good water dispersal characteristics. By contrast, the downslope soil layers showed a limited number of water flow pathways, and these pathways were wider, which resulted in slower water movement. The midslope region primarily served as a zone for water transport (Fig. 12). Consequently, soil water in the upslope infiltrated more readily into deeper soil layers, which effectively replenished groundwater resources (Jarvis, 2020). The downslope experienced slower rates of soil water infiltration, with high SWC in the shallow soil layers (Fig. 12). The soil layer of the downslope was dominated by macropore with high interaction, and the soil matrix flow reached a deeper depth. With the addition of the recharge of soil water from the upslope and the midslope, the shallow soil layer of the

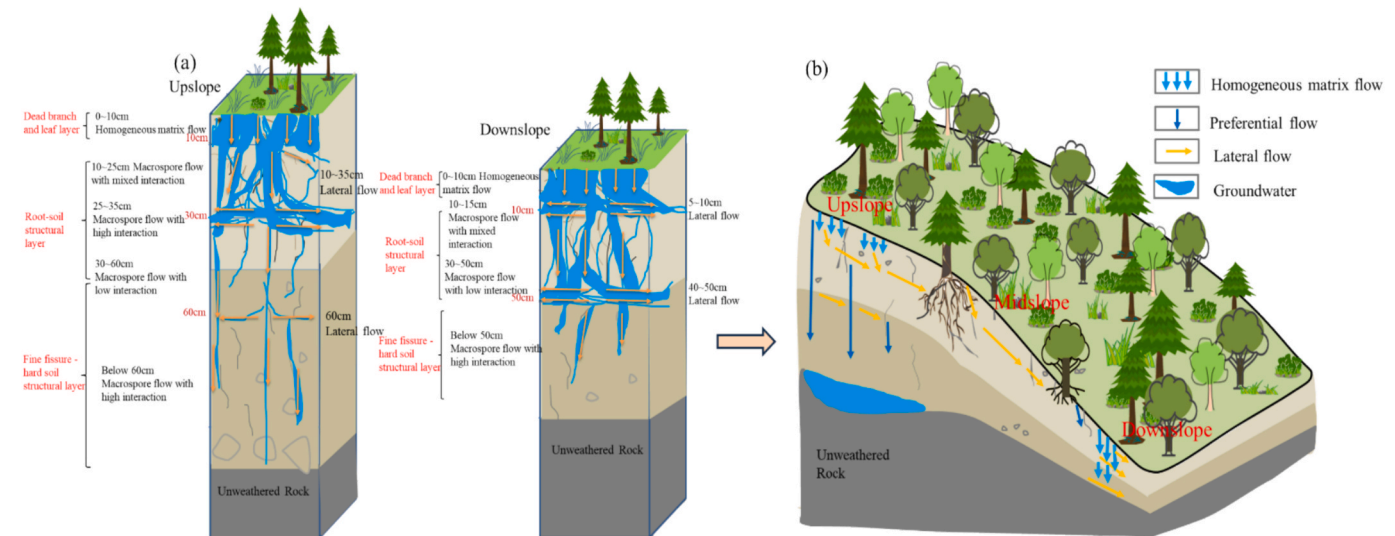


Fig. 12. Conceptual framework of soil water transport paths in the study area ((a) Schematic soil columns at upslope and downslope positions, based on dye tracer patterns, soil profile observations, and measured soil properties. (b) Simulated water flow pathways along the hillslope inferred from (a).).

downslope was prone to become saturated and then generated runoff. These findings were inconsistent with the water transport pattern in the purple soil hilly area, as illustrated in our previous studies (Liu et al., 2021). The main rock formation in the Purple Clay Hills area constituted a shallow sand–mudstone interactive aquifer formation group (rock formation with many fissures). Moisture was easily transported through the soil layer into the rock layer, and then water transport occurred. However, the bedrock in this study area was mainly andesite, which did not easily produce fissures, so moisture transport in the short term was mainly concentrated in the soil layer. Therefore, the rate of runoff production was faster in the forested hilly andesitic critical zones, particularly for those under the subtropical monsoon climate zone with distinct rainy and dry seasons. Such phenomenon further acted on the patterns and processes of flow production in surface runoff and subsurface runoff (Guo and Lin, 2018).

The study area was located in a forest hilly critical zone with andesite bedrock, which was characterized by low elevation differences and strong soil conservation capabilities. These conditions led to the development of soil profiles with similar content of fine particles across the slope. Additionally, the subcatchment was positioned at the outlet of a larger watershed (Fig. 1), which resulted in the accumulation of sediments. The sedimentation processes, combined with the distribution of andesite bedrock (Gu et al., 2018), contributed to the formation of deeper soil profiles in the upslope. These unique characteristics of the study catchment were significant for hydrological modeling and catchment management. Specifically, the soil profile structure across the hillslope must be considered in hydrological models because the vertical and horizontal percolation were intertwined (Fig. 9). By incorporating regional variability in infiltration processes, dynamic water changes at different locations and during various hydrological cycle phases can be more effectively modeled (Jia et al., 2024). The consideration of lateral preferential flow can also improve the initial response in the simulated water arrival time (Chen et al., 2024). This research provided new insights into the mechanisms governing soil water flow and transport dynamics in hillslopes, particularly when modeling water flow paths in hilly andesite areas.

5. Limitations of the research

A double-ring infiltration method was employed in the study, and pure water was added to the outer ring to simulate a wet soil environment during the rainy season for the dye tracer of infiltration, but the

pure water in the outer ring might influence the paths of the dye tracer and affect the experimental results. To address the limitation, future studies could consider using the single-ring infiltration method (Jacka et al., 2018, 2021) to avoid uncertainties.

This study addressed the distribution characteristics of soil water movement paths during the heavy rainy pattern in forest hilly andesitic critical zone. High pressure head and large rainfall simulations increased macropore flow to some extent. In addition, due to the natural variability in soil properties across hillslope locations, further infiltration experiments (using other tracers) and runoff observations in additional sites, along with different rainfall patterns, are required for a more comprehensive understanding of water flow in this critical zone.

6. Conclusions

The objectives of the experimental investigation presented in this article were to characterize qualitatively the soil water pathways on forest hillslopes and underlying factors responsible for SWC variations using infiltration and dye tracer experiments supplemented with soil physicochemical property data.

The number and branching of preferential flows in the upslope were significantly greater than that in the downslope. Dye (i.e., soil moisture) within the upslope infiltrated deeper soil layers, which contributed to groundwater recharge. In the upslope, the predominant flow type was macropore flow with low interaction, whereas the downslope featured macropore flow with high interaction. Soil depth had a direct, important effect on soil water transport, and other soil properties, such as silt content, porosity, bulk density, and wilting point, also directly influenced the water transport paths or indirectly by impacting other factors. Soil layers with numerous flow paths and a uniform distribution of flow path widths, along with a dominance of macropore flow with low interaction, could lead to high SWC levels. Conversely, when the water flow paths were primarily singular or dominated by preferential flow, moisture could not effectively pool in the soil. Analyzing the soil water transport pathways in the hilly critical zone facilitated the understanding of regional hydrological processes, especially those below the surface. Subsurface hydrological pathways were the key to accurate runoff modelling. These findings enhance our understanding of hydrological processes in the hilly andesitic critical zone and contribute to improving model performance in runoff simulations.

CRediT authorship contribution statement

Haowen Liu: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Data curation. **Yanqing Yang:** Writing – review & editing, Supervision, Software, Data curation. **Jiayun Zhang:** Supervision. **Jialiang Tang:** Writing – review & editing, Methodology, Conceptualization. **Aiming Huang:** Investigation. **Aimin Liao:** Supervision, Resources. **Sicheng Wan:** Software, Investigation. **Cuishan Liu:** Writing – original draft. **Guoqing Wang:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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